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A Case for the Use of Chroma Cartesian Colour Representations for Image Classification on Plant-Based Domains

Alexis J S Payne[1] Gail Hopkins[1] Shreyank N Gowda[1] Isaac Triguero[2,3,1]
Michael P Pound[1]
[1]School of Computer Science, University of Nottingham
{alexis.payne, gail.hopkins, shreyank.narayanagowda, michael.pound}@nottingham.ac.uk
[2]Department of Computer Science and Artificial Intelligence, University of Granada
triguero@decsai.ugr.es
[3]DaSCI Andalusian Institute in Data Science and Computational Intelligence, Granada, Spain

Abstract

The primary approach in computer vision across all domains is to work with the Red, Green, Blue (RGB) colour representation. This is most clear in the use of transfer learning, where leveraging known good weights for a model, pretrained on large datasets in RGB, is commonplace. Despite the dominance of RGB, alternative colour representations, such as Hue, Saturation, Value (HSV), have seen use in plant-based domain specific tasks, or with models that are trained from the ground up. In this paper we have explored a wide array of known colour representations, across a variety of plant and plant disease identification datasets. These show that there is not just several colour representations that consistently outperform RGB in training from scratch, but a common construction of the representations that do: an isolated Luminosity channel, similar to a black and white version of the image; combined with a Cartesian representation of the Chroma component, the colour information. We propose a new and effective addition to this class, H2SV, which derives from the relatively widely used HSV representation, improving upon its polar representation of colour information and limited effectiveness when used with neural networks. We conduct extensive experiments that show the effectiveness of these colour representations over RGB is not limited to a single architecture, and this trend is seen across almost all plant datasets. We also show that these benefits can be combined with appropriately pretrained weights so both the computer vision norm of transfer learning, and these alternative colour representations can be leveraged at once. Relevant code tooling is available via PyPI.

1. Introduction

Agriculture currently faces the combined challenges of a growing global population, the threat of climate change, resource depletion, and dietary shifts [1]. Biosciences, plant sciences, and other flora- and fauna-based domains are active and important frontiers of research in addressing these challenges [31]. Accurate, automated plant measurement enables the scale of experimentation necessary to drive genetic discovery, and has become a key tool in these fields.

For many plant measurement problems, machine learning, and computer vision specifically, are well-positioned to help us [20]. However, common machine vision technologies have not been designed with a focus on this domain. Sensors are typically oriented around the human model of vision, in essence capturing images that look accurate to our eye, rather than necessarily the representation that enables the most accurate scientific analysis. Compounding this problem, computer vision and machine learning techniques are often designed and trained around general problems, in which plants are either a small part of datasets, or not included at all. These techniques, therefore, often adapt poorly into specialised domains.

Some technologies have been adopted in science to better address domain-specific issues. The most suited to agriculture and plant analysis are hyper- and multi-spectral sensors, able to extract significantly more information than the human eye can [24]. These suffer from being large, expensive, and slow to operate, significantly limiting the scope and ease of data acquisition [38]. While there are simpler multi-spectral offerings, such as Colour and InfraRed (CIR), these are a compromise between data gathering ability on the one hand and cost and availability on the other. At the extreme of this end of the spectrum are the simple Red, Green and Blue (RGB) sensors that directly mimic the physiology of human vision collecting Red, Green, and Blue

emission peaks. These are ubiquitous in consumer equipment, and so have inevitably become the default choice for most scientific image capture. RGB has now become so prevalent across almost all domains, that it goes unchallenged as the de facto standard colour model for machine learning. This assumes that a colour representation designed to align with our visual system is inherently the best choice for training machine learning models.

Further entrenching the use of RGB as the colour representation of choice is the use of pre-trained networks, both broadly, and within plant-based domains. In particular, the use of pre-training on a general purpose task such as ImageNet [4] has been shown to be very effective in plant-based domains [2]. This is a trend that has been upheld in recent years with the PlantCLEF challenge leaning into pre-training on ImageNet in 2022 [7] and 2023 [8], and while the 2024 edition moved away from this [9] it was only towards the pre-trained, RGB models that the competition organisers provided [10]. This all serves to place RGB at an implicit advantage compared to any other colour representation by virtue of not just the pre-trained weights being created for RGB usage, but also the way that network architectures themselves are designed.

In this work we demonstrate the value of Chroma Cartesian colour representations across a large variety of plant-based classification problems and approaches. Through this we show that the improvements over RGB are statistically significant and can be seen across multiple model architectures and in both from scratch training and with pre-trained networks. Our key contributions are:

- The introduction of H2SV, a new colour representation, which outperforms RGB in 29 out of 33 experiments performed; and to a statistically significant degree in all arrays of experiments.
- The identification of a shared class of Chroma Cartesian colour representations including H2SV, L*u*v*, L*a*b* and the Y-family of colour representations that also outperform RGB.
- Experimentation that shows that the phenomenon of RGB being outperformed is not limited to a single model architecture or training recipe.
- Providing pre-trained weights, trained on ImageNet-1K in H2SV for the backbones used in this paper, along with a library for working with alternative colour representations in PyTorch available on PyPI.

The rest of this paper is organised as follows: Sec. 2 will provide a deeper literature review on image classification, the use of alternative colour space representations in general, and in plant-based domains, as well as the motivation for developing H2SV; Sec. 3 will provide the derivation of the H2SV colour representation; Sec. 4 will describe the datasets and experimental frameworks used and present the results, along with analysis of them; Sec. 5 will summarise our conclusions.

2. Background

This section briefly describes the topics touched on in this paper: the architecture and mechanism of the relevant models in Sec. 2.1; the use of alternative colour representations in general purpose computer vision in Sec. 2.2; the use of alternative colour and spectral representations on plant-based domains in Sec. 2.3; and finally focuses on the specific motivation for creating H2SV in Sec. 2.4.

2.1. Image Classification Models and Architecture

The recent rise of deep learning has fuelled a rapid and continual evolution of neural networks. While these networks have an origin in mimicking single neurons [36], we now see deep networks with parameter counts that can reach one trillion [5]. Simple convolutional networks have evolved to become deeper, and we have seen the introduction of powerful attention mechanisms within vision transformers [16] and hybridisations of transformers and convolutional networks [21]. Regardless of the specifics of any one particular architecture, in practice these models are used for image classification in very consistent ways. They act as a function that maps an input tensor representing the image into a 1-dimensional tensor representing the prediction confidences for each potential class. The image representation is remarkably consistent across architectures and domains, with colour images represented as a 3-dimensional tensor of size C by H by W representing a C-channel image, of height H and width W. The most popular format, RGB, comprises 3 channels representing Red, Green and Blue respectively, and results in a tensor of size 3 by H by W. Other formats, such as greyscale or multi-spectral images, will vary the number of channels.

2.2. Alternative Colour Representations in Computer Vision

Various alternative colour representations exist and see varying degrees of use in different domains. In fact the idea of an infinite number of trichromatic representations has existed for more than 100 years [30]. Those of interest in this paper however are those that present colour differently to RGB, remaining equivalent while being distinct representations. Examples of these, and specifically the Chroma Cartesian class that we demonstrate the value of in this paper, include: YUV, and the Y-family of colour representations, which separate Luminosity from Chroma information as a means to encode black and white and colour television efficiently in parallel [41]; and L*u*v* and L*a*b*, which came from efforts by the International Commission on Illumination (CIE) to achieve perceptual uniformity, i.e. a 1 unit difference feeling like it is of the same size wherever

it is made in the colour space, and both separate the Luminosity, L, from the Chroma information modelled using Cartesian coordinates within the colour space, on the other two values.

Alternatives to RGB, including the above mentioned, do see use in computer vision tasks. A notable work is ColorNet [11], which used a wide variety of colour representations in ensemble for image classification. Beyond competitive performance results, this work showed that training in different colour representations led to models with different performance characteristics, i.e. models trained in different colour spaces were strong and weak on different classes of the dataset.

Work has also shown the value of creating domain-specific colour representations [37]. Here specific cell stains are isolated into specific colour channels for histochemical analysis, resulting in a robust method of sperating stains by transforming data gathered using RGB sensors.

More recently work on enhancing underwater images has made use of the Hue, Saturation, and Value (HSV) colour representation in parallel to RGB [28]. Here the richer chromatic data representation that HSV offers benefits the final, enhanced image. HSV specifically has also generated interest in food science, where it has been leveraged for quantifying food quality where colour is important [43]. Even more directly, Colour-analyzer [22] used both HSV and L*a*b* to handle colour-based pixel analysis with more accuracy than using RGB would provide. Nain *et al*. [33] did work in a similar vein to what this paper presents, with promising results, but was far more limited in scope looking at only one dataset, and one non-standard architecture.

2.3. Plant-Based Domains and Spectral Information

Plant-based imaging benefits greatly from spectral information, whether remote [38] or proximate [23]. These hyper- or multi-spectral approaches leverage domain features such as the way that plants absorb and reflect specific wavelengths of light to extract information that is not found in a simple RGB or CIR image. But even when the more limited spectra are used, they are often processed to derive more specific information through the use of indexes such as the normalised difference vegetation index [34] or leaf rust disease severity indices [27]. These effectively embed expert knowledge in how to understand the spectral information. HSV also sees use in this domain, and can be desirable when working with the more limited spectral information provided by RGB sensors. This can have value in the analysis of singular pixels [19]. In remote sensing, Luus *et al*. [25] found that, while effective, HSV was eventually outperformed by RGB, indicating that training effectively on the polar coordinate system of HSV remains a challenge.

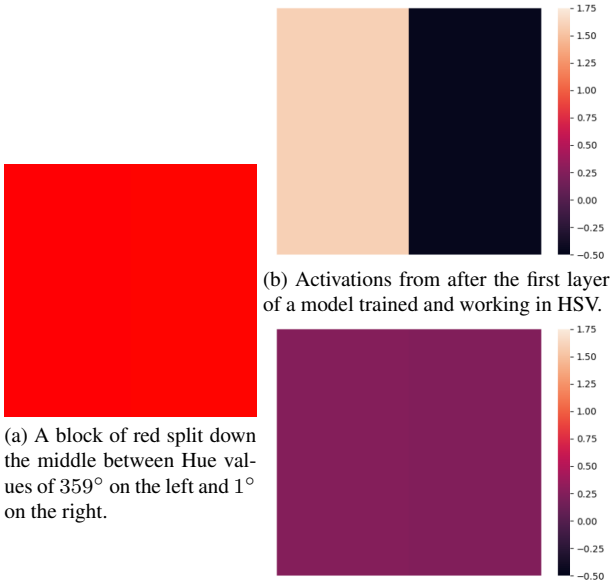
2.4. Motivation for an HSV-Derived Representation

A small proportion of papers [15] that have worked with alternative colour representations on agricultural datasets, focus on HSV. This is a representation that similar to L*u*v and L*a*b* is more motivated by the psychological perception of colour, rather than the physiological [6]. For another example, the work of Nain *et al*. [33] demonstrated good performance of HSL, a close relative of HSV, performing well on a specific dataset and architecture.

In HSV, Hue represents the colour as a cycle of red, through blue and green and back to red. Saturation represents the intensity of colour as distance from an origin of grey, and Value the brightness. Separating the colour from the intensity and brightness intuitively allows us to develop algorithms that separately focus on each aspect of colour. For example, a consistent Hue with varied Saturation and Value may better reflect an object under different lighting conditions [28]. Though HSV does see some use in classification, it has seen limited uptake, and RGB is often found to outperform it [25]. In the domain of plant science, HSV has shown benefits in the identification of plant features, motivating its use in preprocessing methods [19] prior to the use of RGB for the final classification. It has also found use in tools that operate on single pixels of an image at a time to perform analysis without using a deep learning model such as in Colour-analyzer [22]. Despite this, HSV has not been widely used in most plant-based computer vision domains, which remain dominated by RGB representations.

A potential reason for the limited uptake of HSV in machine learning models is the polar coordinate nature of the representation of Hue, which varies from 0° to 360°. The cyclic nature of this value is ill-suited for use in mathematical operations. By way of example, Fig. 1 shows how two very similar tones are interpreted very differently by a convolutional network. Hue values that are close to this point of discontinuity can take radically different values, even while being near indistinguishable to the human eye and representing the same object or feature. These discontinuities will produces artefacts in activations within neural networks, that may interfere with model training. Polar coordinates are further complicated by the fact that an x° change can have different meanings in different points of a wheel, e.g. both positive and negative change could mean a change of a colour towards being more green, or more red. The value of H also becomes less meaningful depending on the Saturation, where a low Saturation, S, allows Hue to vary without a perceptual change in the output image. In the most extreme case, when Saturation reaches zero, Hue is arguably undefined.

Despite these limitations, we argue that HSV has the potential to be a more useful model in specific domains, such as plant-based tasks. However, it will require effort to make Hue more compatible with the way convolutional networks



(a) A block of red split down the middle between Hue values of 359° on the left and 1° on the right.

(b) Activations from after the first layer of a model trained and working in HSV.

(c) Activations from after the first layer of a model trained and working in H2SV, presented on the same reference scale as Fig. 1b.

Figure 1. An example of the artefacts common to HSV that H2SV solves.

are trained. In this paper we present H2SV, a new colour representation inspired by HSV but designed to overcome its limitations. We separate the H channel of HSV into 2 components, each presented in their own channel, creating a 4-channel representation. By using 2 channels for H, distances between colours are consistent, predictable, and continuous regardless of the position on the colour wheel.

3. H2SV: An Evolution of the HSV Colour Representation

In this section we present the derivation of the H2SV colour representation, which we designed to be a straightforward Chroma Cartesian replacement to HSV that avoids the polar coordinates that hinder model performance.

The full derivation of the HSV values from RGB ones can be seen in Eqs. (1) to (7). The mapping from RGB to HSV is non-linear, first involving the calculation of a number of intermediate values representing the relative levels of the R, G, and B channels, shown in Eqs. (1) to (3). Here the R, G, and B values are taken as being in the interval $[0, 1]$

$$M = \max(R, G, B) \tag{1}$$

$$m = \min(R, G, B) \tag{2}$$

$$C = \text{range}(R, G, B) = M - m \tag{3}$$

We then calculate the intermediate H' value via a piecewise function shown in Eq. (4). Here each of these cases covers a third of the colour wheel. Specifically in instances where $C = 0$, H' is canonically undefined and as such we take 0 as a reasonable default.

$$H' = \begin{cases} 0 & C = 0 \\ \frac{G-B}{C} \mod 6 & M = R \\ \frac{B-R}{C} + 2 & M = G \\ \frac{R-G}{C} + 4 & M = B \end{cases} \tag{4}$$

At this stage H' exists in the interval $[0, 6[$. We could then scale to produce the canonical colour representation H as a value that varies between 0° and 360°. However, in computer vision, it is common practice to scale all values to the interval $[0, 1[$, and so H' is scaled using a division by 6, shown in Eq. (5).

$$H = \frac{H'}{6} \tag{5}$$

S is calculated as the ratio of range C and the maximum, M. This is achieved using a piecewise function, shown in Eq. (6) to prevent division by 0. V is simply M, or the maximum value of R, G, and B, as seen in Eq. (7).

$$S = \begin{cases} 0 & M = 0 \\ \frac{C}{M} & \text{otherwise} \end{cases} \tag{6}$$

$$V = \max(R, G, B) = M \tag{7}$$

It is important to note here that the H value is angular and thus cyclic. This means that for fixed values of S and V, varying H will describe a circle with a point of discontinuity between the maximum and minimum values of H. The shape described can be seen in Fig. 2, where each value of H will produce a point on this circle. With H always being measured clockwise from the positive y half axis, the discontinuity occurs where it crosses this positive y half axis. We can also observe that each of those points can be uniquely defined by Cartesian coordinates, i.e. x and y values. We therefore propose to replace the H value with values H_1 and H_2, representing these x and y values, and for which no point of discontinuity exists.

We derive H_1 and H_2 as shown in Eqs. (8) and (9), as a unit vector representing the direction of H. These are combined with the S and V values to form a new 4-channel H2SV representation (H_1, H_2, S, V).

$$H_1 = sin(H) \tag{8}$$

$$H_2 = cos(H) \tag{9}$$

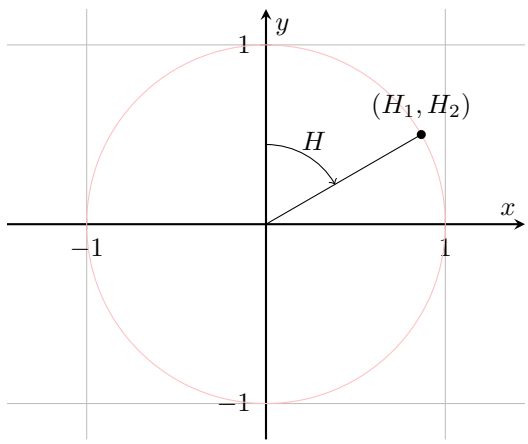


Figure 2. An example plot of a circle, such as those described by varying H with fixed S and V values of 1.

4. Experimental Study

This section will present the datasets considered, Sec. 4.1; the experimental frameworks used, Sec. 4.2; and discuss the results obtained, Sec. 4.3.

4.1. Datasets and Metrics

We perform experiments across 11 popular, plant-based classification datasets to support the hypothesis that Chroma Cartesian alternatives to RGB are valuable in this domain. These datasets cover a variety of classification tasks common to plant science: leaf sample classification, plant disease classification, and plant species identification. Further detail can be found in Tab. 1, with representative images in Fig. 3.

Ex situ, i.e. controlled environment, leaf sample datasets including Swedish Leaf [39], Flavia [42], Leafsnap [18], Folio [29], D-Leaf [40], PlantaeK [17] were chosen as baseline challenges as well as some classic benchmarks in the wider domain. By contrast, in situ datasets such as Cassava [32] offer the additional challenge of plants being photographed in uneven light and varied backgrounds, i.e. in the context of their natural environment. In addition to the leaf sample datasets which classify by species, we looked at a selection of plant disease datasets such as PlantVillage [14], Citrus Leaves (and Fruit) [35], Cassava [32], and Plant Pathology [3]. In these there may be one or multiple species. Where multiple species exist we naïvely treat each species and health condition pairing, including healthy, as a distinct and unrelated class.

With these datasets being mostly balanced across classes, we present top-1 accuracy as the key performance metric. In datasets with more than 5 classes, top-5 accuracy is also reported. To be explicit, top-n accuracy reflects if the most confident n predictions of the classifier include the ground truth; i.e. top-1 accuracy reflects if the classifier is "right".

We performed statistical analysis on the results, which can be found in Tabs. 3, 5 and 7. Here we performed an $N \times N$ analysis, first producing a ranking using the Friedman procedure based on the top-1 accuracies of the experiments, and we then used Holm post hoc analysis. An $N \times N$, in this case a 6×6, comparison compares all of the methods, in this case colour representations, against all other methods, as opposed to a $1 \times N$ comparison that would only compare the highest ranked to all of the others. The result of this is p-values for the null-hypothesis, that the higher ranked, by the Friedman procedure, colour representation is not different to the colour representation it is being compared to. For considering these we set $\alpha = 0.05$ as our threshold for significance and have bolded cells of the table where the null hypothesis can be rejected given this threshold. Meaningless comparisons, i.e. between a representation and itself, or between a representation and one that was higher ranked by the Friedman procedure, are omitted with '-'.

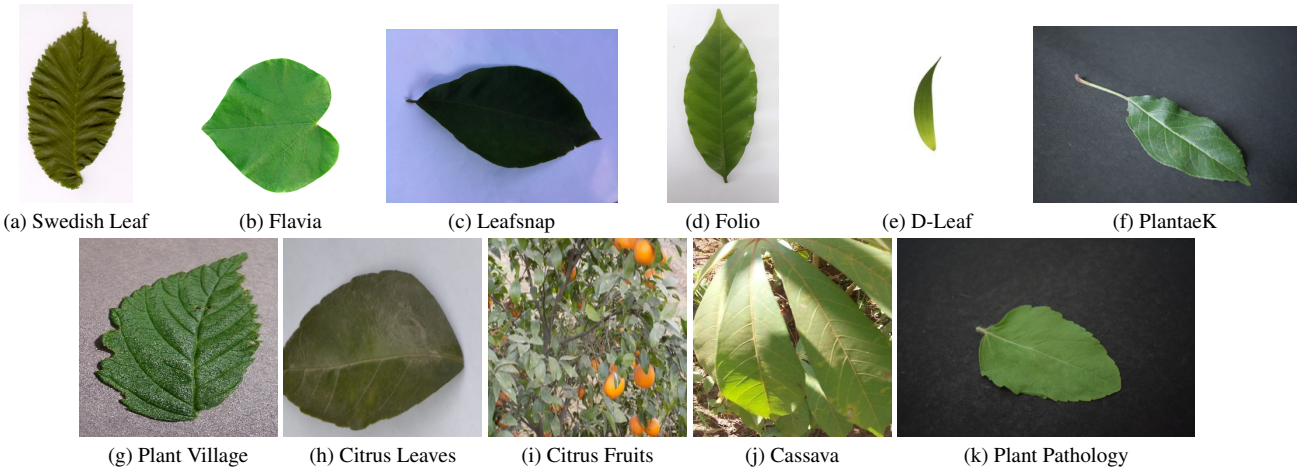
Preprocessing is performed across all images, with some additional consideration of consistency across colour spaces. In all cases we begin by resizing images to 236 by 236 pixels, then taking a central crop of 224 by 224 pixels. This is done while an image still exists in RGB because the interpolation techniques used in resizing an image, specifically bilinear interpolation, are designed with RGB in mind and can produce unexpected results in alternative colour representations. Equally, normalisation should not be performed before this or before transforming to an alternative colour representation as this moves the RGB values out of their usual ranges. Images are then transformed into an alternative colour representation, with this process varying based on the representation in question. This may occasionally require use of an intermediate representation, usually XYZ. Finally, all images are normalised; this is left until last to allow the other processes to occur as expected. This is performed with reference to the channel-wise dataset means and standard deviations, calculated in each final colour representation. In our experiments these metrics were calculated on the training set, converted into the relevant colour representation, without any resizing or cropping. This was done per dataset, resulting in normalisation specific to the dataset.

4.2. Experimental Designs

Initial experiments use the ConvNeXt Tiny architecture [21] as a contemporary backbone, and are trained from scratch as this was the most effective way to work in a number of colour representations in a reasonable time frame. These experiments consisted of a broad selection of the datasets described in Sec. 4.1 across a wide variety of colour repre-

Table 1. Properties of Datasets Used

Dataset	# Samples	# Classes	Resolution	Environment	Subject	Classification
Swedish Leaf [39]	1,410	15	Variable	Ex situ	Leaves	Species
Flavia [42]	1,907	32	800 by 600	Ex situ	Leaves	Species
Leafsnap [18]	30,866	185	Variable	Ex situ	Leaves	Species
Folio [29]	640	32	1,980 by 1,024	Ex situ	Leaves	Species
D-Leaf [40]	1,632	43	Variable	Ex situ	Leaves	Species
PlantaeK [17]	2,153	16	6,000 by 4,000	Ex situ	Leaves	Species
PlantVillage [14]	54,303	38	Variable	Ex situ	Leaves	Species, Healthy/Specific Disease
Citrus Leaves [35]	609	5	256 by 256	Ex situ	Leaves	Healthy/Specific Disease
Citrus Fruits [35]	150	5	256 by 256	Mixed	Fruits/Fruit Wide Shots	Healthy/Specific Disease
Cassava [32]	5,656	5	Variable	In situ	Leaves	Healthy/Specific Disease
Plant Pathology [3]	4,503	22	Variable	Ex situ	Leaves	Species, Healthy/Diseased



(a) Swedish Leaf (b) Flavia (c) Leafsnap (d) Folio (e) D-Leaf (f) PlantaeK

(g) Plant Village (h) Citrus Leaves (i) Citrus Fruits (j) Cassava (k) Plant Pathology

Figure 3. Example Images From the Datasets Used.

sentations that were then reduced down to the representations of interest shown in this paper. The full selection of results can be found in the supplementary materials. Notably choices were made such as: YUV was chosen to represent the family that includes YC_BC_R, YP_BP_R, YD_BD_R, and YIQ, as these are scalar multiplications and linear mappings of each other, and particularly after normalisation we found that results in these colour spaces did not exhibit significantly different performance; HSV was chosen for use and HSL and HSI not considered for the same reason; none of the cell staining colour representations proposed by [37] were brought forward due to them producing poor results on our array of datasets; RGBCIE and XYZ were similarly not brought forward due to results that were either not significantly different from, or significantly worse than RGB, respectively.

Being trained from scratch, the weights of these models are initialised randomly, they are then trained for 100 epochs on the given dataset using cross entropy loss and the Adam optimiser with a learning rate of 0.001. Additionally, these experiments were performed under 5-fold cross validation, consisting of five distinct experiments where models are trained on 4 folds and tested on the 5th. The results that are then presented being the average across the 5 runs. These initial results, filtered for the colour representations of interest, are presented in Tab. 2.

In order to show that this phenomenon, where Chroma Cartesian colour representations are providing performance gains, was not limited to the ConvNeXt Tiny architecture, we use ResNet-50 [13] for the next set of experiments. Specifically the V1.5 version proposed by Nvidia and adopted by PyTorch based on the bottleneck modifications proposed by He *et al.*, the originators of ResNet [12]. These were trained using the same 5-fold cross validation, cross entropy loss, Adam optimiser and 0.001 learning rate, but given 30 epochs of training. The results, again an average of the five runs, are presented in Tab. 4.

The final array of tests use the same ResNet-50 architecture, but are initialised not with random weights as with the training from scratch but from weights pre-trained on

ImageNet-1k [4]. The goal of these was to demonstrate that the value of alternative colour representations was not limited to from scratch training and the results are presented in Tab. 6. ImageNet was chosen as the pre-training dataset instead of a plant-based example based on the results of Carranza-Rojas *et al*. [2]. The hyperparameters of this experiment were the same as for training ResNet-50 from scratch, but only 15 epochs were given. The backbones themselves were trained based on the reference recipes provided by PyTorch [26] which are used for training the pre-trained backbones provided by torchvision. The one adjustment made to these recipes was the change from from 8 to 4 GPUs for training, which was made due to the limited resources available to us. Because of this change to the training recipe, the RGB backbone was also trained by us rather than the reference weights being used.

4.3. Results and Discussion

The first results we will present are a select set from the first experimental array on ConvNeXt Tiny, found in Tab. 2. The full table can be found in the supplementary materials, including predominantly weak results from other canonical colour representations, and the statistically similar collection of results from the Y-family of colour representations, which are instead flattened to just reporting the YUV results. Results for top-5 accuracy are omitted with "-" where 5 classes or fewer occur.

In these results we can see H2SV performs well on many experiments, particularly compared to HSV which it is designed to improve upon. RGB can also be seen to be consistently poor, with its best relative performance being on the very saturated PlantVillage dataset in which the margins between approaches are very narrow. Our analysis of these results seen in Tab. 3 demonstrates that H2SV, YUV, L*u*v*, and L*a*b* form a class where the null hypothesis, that there is no difference between distributions, cannot be rejected, but they can all reject the null hypothesis when compared to HSV and RGB.

Presented in Tab. 4 the results for this same array of colour representations on ResNet-50 shows many of the same trends as ConvNeXt Tiny. The notable differences are in the PlantVillage results where the saturated results now pivot away from RGB, and in the statistical analysis found in Tab. 5. Here the Chroma Cartesian colour representations, H2SV, L*a*b*, YUV, L*u*v*, can still be shown to be outperforming RGB in a statistically significant way, and are not significantly separable from each other. They are however no longer statistically separable from HSV despite broadly outperforming it. This supports our hypothesis that Chroma Cartesian colour representations outperforming RGB is not a phenomenon limited to the ConvNeXt architecture.

Finally, Tab. 6 presents the results of the experiments with pre-training. They continue to show the trend of RGB being outperformed by Chroma Cartesian colour representations. Through not necessarily outperforming the results seen in Tab. 4 in which twice as many epochs were given on the target dataset. We find there are two big takeaways from this data: firstly, that transfer learning of this type is not limited to working in RGB and can be incredibly effective in alternative colour representations; secondly, that, as shown in Tab. 7 Chroma Cartesian colour representations all still outperform RGB to a statistically significant degree.

There are some downsides to working with alternative colour representations in these ways. H2SV, when used exclusively at runtime via transforms, requires around 33% more memory than 3-channel representations, and would do in storage if used there. This is specific to the images being held in memory before being passed to the GPU and the model for training. In the model itself, the increased memory cost does not propagate through the network and so the GPU memory need and trained model size are negligibly larger. Training backbone models in a new colour

Table 2. Accuracies by Dataset and Colour Representation for ConvNeXt Tiny Trained From Scratch

Dataset	Accuracy	RGB	H2SV	HSV	YUV	L*a*b*	L*u*v*
Swedish	Top-1	0.9124	**0.9439**	0.9217	0.9285	0.9260	0.9378
	Top-5	0.9983	0.9974	0.9983	**1.0000**	0.9984	0.9982
Flavia	Top-1	0.8904	**0.9352**	0.8794	0.9108	0.9047	0.9135
	Top-5	0.9932	**0.9971**	0.9951	0.9942	0.9925	0.9938
Leafsnap	Top-1	0.8606	**0.8801**	0.8724	0.8753	0.8769	0.8679
	Top-5	0.9729	**0.9759**	0.9731	0.9753	0.9755	0.9749
Folio	Top-1	0.7566	**0.8505**	0.8093	0.8115	0.7872	0.8179
	Top-5	0.9472	**0.9788**	0.9674	0.9538	0.9633	0.9604
D-Leaf	Top-1	0.5501	0.6311	0.6154	0.6841	0.6745	**0.6863**
	Top-5	0.7810	0.8296	0.8076	0.8277	**0.8351**	0.8274
PlantaeK	Top-1	0.6154	0.6199	0.5931	0.6180	0.6193	**0.6273**
	Top-5	0.9680	0.9552	0.9589	**0.9704**	0.9634	0.9694
PlantVillage	Top-1	**0.9722**	0.9612	0.9661	0.9681	0.9632	0.9709
	Top-5	**0.9991**	0.9976	0.9986	0.9986	0.9984	0.9986
Citrus Leaves	Top-1	0.8851	0.8914	0.8680	**0.9038**	0.8902	0.8952
	Top-5	-	-	-	-	-	-
Citrus Fruits	Top-1	0.4878	0.5260	0.5294	**0.5931**	0.5629	0.5175
	Top-5	-	-	-	-	-	-
Cassava	Top-1	0.3602	**0.3692**	0.3553	0.3526	0.3667	0.3554
	Top-5	-	-	-	-	-	-
Plant Pathology	Top-1	0.8752	**0.8988**	0.8505	0.8781	0.8875	0.8856
	Top-5	0.9870	**0.9941**	0.9876	0.9895	0.9905	0.9906

Table 3. Unadjusted P-values for Rejecting the Null-Hypothesis for ConvNeXt Tiny Trained From Scratch Under Friedman Ranking and Holm Post Hoc Analysis

Hypothesised "Better" Colour Representation	H2SV	L*u*v*	YUV	L*a*b*	HSV	RGB
H2SV	-	0.6485	0.3619	0.2100	**0.0009**	**0.0009**
L*u*v*	-	-	0.6485	0.4250	**0.0044**	**0.0044**
YUV	-	-	-	0.7324	**0.0167**	**0.0167**
L*a*b*	-	-	-	-	**0.0402**	**0.0402**
HSV	-	-	-	-	-	1.0000
RGB	-	-	-	-	-	-

Table 4. Accuracies by Dataset and Colour Representation for ResNet-50 Trained From Scratch

Dataset	Accuracy	RGB	H2SV	HSV	YUV	L*a*b*	L*u*v*
Swedish	Top-1	**0.9527**	0.9288	0.8072	0.9512	0.9471	0.9124
	Top-5	0.9981	**1.0000**	0.9913	0.9975	0.9993	0.9964
Flavia	Top-1	0.8366	0.9067	0.9072	0.9061	0.8371	**0.9313**
	Top-5	0.9800	0.9930	0.9935	0.9919	0.9607	**0.9945**
Leafsnap	Top-1	0.8638	**0.8911**	0.8864	0.8823	0.8840	0.8765
	Top-5	0.9808	**0.9866**	0.9846	0.9862	0.9860	0.9854
Folio	Top-1	0.6960	**0.8079**	0.7801	0.7805	0.7003	0.7865
	Top-5	0.9022	**0.9807**	0.9693	0.9670	0.9361	0.9663
D-Leaf	Top-1	0.3658	**0.4899**	0.3948	0.3261	0.2887	0.3934
	Top-5	0.6437	**0.7661**	0.6768	0.5369	0.5477	0.6617
PlantaeK	Top-1	0.6556	0.7238	0.6605	0.7231	**0.7432**	0.7296
	Top-5	0.9792	0.9760	0.9887	**0.9950**	0.9863	0.9928
PlantVillage	Top-1	0.9758	0.9792	0.9718	0.9807	**0.9819**	0.9773
	Top-5	0.9993	0.9991	0.9960	0.9994	**0.9996**	0.9991
Citrus Leaves	Top-1	0.7217	0.8735	0.9117	**0.9226**	0.9110	0.8838
	Top-5	-	-	-	-	-	-
Citrus Fruits	Top-1	0.5415	0.5475	**0.6333**	0.6023	0.5689	0.5417
	Top-5	-	-	-	-	-	-
Cassava	Top-1	0.4032	0.4492	0.4382	0.4496	**0.4512**	0.4303
	Top-5	-	-	-	-	-	-
Plant Pathology	Top-1	0.8649	0.8750	0.8576	0.8624	0.8900	**0.8924**
	Top-5	0.9894	0.9916	0.9908	0.9859	0.9904	**0.9943**

Table 5. Unadjusted P-values for Rejecting the Null-Hypothesis for ResNet-50 Trained From Scratch Under Friedman Ranking and Holm Post Hoc Analysis

Hypothesised "Better" Colour Representation	H2SV	L*a*b*	YUV	L*u*v*	HSV	RGB
H2SV	-	0.8197	0.7324	0.4941	0.3051	**0.0044**
L*a*b*	-	-	0.9093	0.6485	0.4250	**0.0088**
YUV	-	-	-	0.7324	0.4941	**0.0122**
L*u*v*	-	-	-	-	0.7324	**0.0304**
HSV	-	-	-	-	-	0.0682
RGB	-	-	-	-	-	-

Table 6. Accuracies by Dataset and Colour Representation for ResNet-50 Fine-Tuning

Dataset	Accuracy	RGB	H2SV	HSV	YUV	L*a*b*	L*u*v*
Swedish	Top-1	0.8332	**0.9047**	0.8572	0.8567	0.8770	0.8897
	Top-5	0.9953	0.9984	0.9980	0.9949	**0.9985**	0.9967
Flavia	Top-1	0.7533	0.8799	0.8579	0.8718	0.7947	**0.8879**
	Top-5	0.9094	0.9882	0.9835	0.9875	0.9586	**0.9959**
Leafsnap	Top-1	0.8087	**0.8406**	0.8263	0.8370	0.8336	0.8141
	Top-5	0.9735	**0.9818**	0.9780	0.9803	0.9801	0.9732
Folio	Top-1	0.5506	**0.7441**	0.6429	0.6181	0.6250	0.7210
	Top-5	0.8584	**0.9544**	0.8983	0.8990	0.9398	0.9470
D-Leaf	Top-1	0.2906	0.3516	0.2737	**0.3636**	0.1779	0.3018
	Top-5	0.6064	**0.7009**	0.5690	0.6728	0.3758	0.5542
PlantaeK	Top-1	0.6045	**0.6867**	0.5533	0.6726	0.6810	0.6848
	Top-5	0.9887	0.9910	0.9753	**0.9938**	0.9882	0.9892
PlantVillage	Top-1	0.9700	0.9689	0.9737	0.9688	0.9709	**0.9749**
	Top-5	0.9989	**0.9995**	0.9993	0.9979	0.9992	0.9988
Citrus Leaves	Top-1	0.6618	0.9075	0.8351	**0.9323**	0.9106	0.9008
	Top-5	-	-	-	-	-	-
Citrus Fruits	Top-1	0.4500	0.4287	0.5132	0.5865	**0.5985**	0.5809
	Top-5	-	-	-	-	-	-
Cassava	Top-1	0.4036	0.4600	0.4404	**0.4759**	0.4310	0.4253
	Top-5	-	-	-	-	-	-
Plant Pathology	Top-1	0.8102	0.8466	0.8126	**0.8658**	0.8212	0.8504
	Top-5	0.9838	0.9904	0.9853	**0.9925**	0.9896	0.9920

Table 7. Unadjusted P-values for Rejecting the Null-Hypothesis for ResNet-50 Fine-Tuned After Pre-Training Under Friedman Ranking and Holm Post Hoc Analysis

Hypothesised "Better" Colour Representation	H2SV	L*u*v*	YUV	L*a*v*	HSV	RGB
H2SV	-	0.7324	0.6485	0.2100	**0.0402**	**0.0002**
L*u*v*	-	-	0.9093	0.3619	0.0874	**0.0006**
YUV	-	-	-	0.4250	0.1106	**0.0009**
L*a*b*	-	-	-	-	0.4250	**0.0122**
HSV	-	-	-	-	-	0.0874
RGB	-	-	-	-	-	-

representation is not trivial, these models require significant time and resources to train, it is however very tenable in the context of a field of research and the reusability of these backbones. For example the ResNet-50 backbones train in 350 GPU hours on Nvidia Titan Xp GPUs, likely less on newer GPUs with similar, moderate VRAM requirements.

5. Concluding Remarks

In this work we have provided a case for an entire class of Chroma Cartesian colour representations outperforming RGB on plant-based image classification tasks. We have evidenced this across a large number of datasets using two very different architectures in ConvNeXt Tiny and ResNet-50. We have also shown results are similar when training from scratch or using pretrained weights. We have introduced and shown the value of H2SV, a new Chroma Cartesian colour representation derived from HSV, which is performing strongly within the class and regularly outperformed RGB during our tests. We argue here that there is a strong case for the use H2SV on plant-based domains for image classification work, and that the value of Chroma Cartesian colour representations in other computer vision tasks merits further exploration.

Our future work will consider the application of H2SV and other Chroma Cartesian colour spaces to a wider variety of architectures and problems, including vision transformers and segmentation tasks.

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References

[1] Lefteris Benos, Aristotelis C Tagarakis, Georgios Dolias, Remigio Berruto, Dimitrios Kateris, and Dionysis Bochtis. Machine learning in agriculture: A comprehensive updated review. *Sensors*, 21(11):3758, 2021. 1

[2] Jose Carranza-Rojas, Herve Goeau, Pierre Bonnet, Erick Mata-Montero, and Alexis Joly. Going deeper in the automated identification of herbarium specimens. *BMC evolutionary biology*, 17:1–14, 2017. 2, 7

[3] Siddharth Singh Chouhan, Uday Pratap Singh, Ajay Kaul, and Sanjeev Jain. A data repository of leaf images: Practice towards plant conservation with plant pathology. In *2019 4th International Conference on Information Systems and Computer Networks (ISCON)*, pages 700–707. IEEE, 2019. 5, 6

[4] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pages 248–255. Ieee, 2009. 2, 7

[5] William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022. 2

[6] Valensi Georges. System of television in colors, 1945. US Patent 2,375,966. 3

[7] Hervé Goëau, Pierre Bonnet, and Alexis Joly. Overview of plantclef 2022: Image-based plant identification at global scale. CEUR-WS, 2022. 2

[8] Hervé Goëau, Pierre Bonnet, and Alexis Joly. Overview of plantclef 2023: image-based plant identification at global scale. In *CLEF 2023 Working Notes-24th Conference and Labs of the Evaluation Forum*, pages 1972–1981, 2023. 2

[9] Hervé Goëau, Vincent Espitalier, Pierre Bonnet, and Alexis Joly. Overview of plantclef 2024: multi-species plant identification in vegetation plot images. CEUR-WS, 2024. 2

[10] H Goëau, JC Lombardo, A Affouard, V Espitalier, P Bonnet, and A Joly. Plantclef 2024 pretrained models on the flora of the south western europe based on a subset of pl@ ntnet collaborative images and a vit base patch 14 dinov2 (2024). *URL: https://doi.org/10.5281/zenodo.10848262*, 10848263, 2024. 2

[11] Shreyank N Gowda and Chun Yuan. Colornet: Investigating the importance of color spaces for image classification. In *Computer Vision–ACCV 2018: 14th Asian Conference on Computer Vision, Perth, Australia, December 2–6, 2018, Revised Selected Papers, Part IV 14*, pages 581–596. Springer, 2019. 3

[12] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Delving deep into rectifiers: Surpassing human-level performance on imagenet classification. In *Proceedings of the IEEE international conference on computer vision*, pages 1026–1034, 2015. 6

[13] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016. 6

[14] David Hughes, Marcel Salathé, et al. An open access repository of images on plant health to enable the development of mobile disease diagnostics. *arXiv preprint arXiv:1511.08060*, 2015. 5, 6

[15] Andreas Kamilaris and Francesc X Prenafeta-Boldú. Deep learning in agriculture: A survey. *Computers and electronics in agriculture*, 147:70–90, 2018. 3

[16] Salman Khan, Muzammal Naseer, Munawar Hayat, Syed Waqas Zamir, Fahad Shahbaz Khan, and Mubarak Shah. Transformers in vision: A survey. *ACM computing surveys (CSUR)*, 54(10s):1–41, 2022. 2

[17] Vippon Preet Kour and Sakshi Arora. PlantaeK: A leaf database of native plants of jammu and kashmir. In *Recent Innovations in Computing: Proceedings of ICRIC 2021, Volume 1*, pages 359–368. Springer, 2022. 5, 6

[18] Neeraj Kumar, Peter N Belhumeur, Arijit Biswas, David W Jacobs, W John Kress, Ida C Lopez, and João VB Soares. Leafsnap: A computer vision system for automatic plant species identification. In *Computer Vision–ECCV 2012: 12th European Conference on Computer Vision, Florence, Italy, October 7-13, 2012, Proceedings, Part II 12*, pages 502–516. Springer, 2012. 5, 6

[19] Sue Han Lee, Chee Seng Chan, Paul Wilkin, and Paolo Remagnino. Deep-plant: Plant identification with convolutional neural networks. In *2015 IEEE international conference on image processing (ICIP)*, pages 452–456. IEEE, 2015. 3

[20] Konstantinos G Liakos, Patrizia Busato, Dimitrios Moshou, Simon Pearson, and Dionysis Bochtis. Machine learning in agriculture: A review. *Sensors*, 18(8):2674, 2018. 1

[21] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A convnet for the 2020s. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11976–11986, 2022. 2, 5

[22] Mackenzie Eli William Loranger, Winfield Yim, Vittorio Accomazzi, Nadia Morales-Lizcano, Wolfgang Moeder, and Keiko Yoshioka. Colour-analyzer: a new dual colour model-based imaging tool to quantify plant disease. *Plant Methods*, 20(1):60, 2024. 3

[23] Amy Lowe, Nicola Harrison, and Andrew P French. Hyperspectral image analysis techniques for the detection and classification of the early onset of plant disease and stress. *Plant methods*, 13(1):80, 2017. 3

[24] Bing Lu, Phuong D Dao, Jiangui Liu, Yuhong He, and Jiali Shang. Recent advances of hyperspectral imaging technology and applications in agriculture. *Remote Sensing*, 12(16): 2659, 2020. 1

[25] Francois PS Luus, Brian P Salmon, Frans Van den Bergh, and Bodhaswar Tikanath Jugpershad Maharaj. Multiview deep learning for land-use classification. *IEEE Geoscience and Remote Sensing Letters*, 12(12):2448–2452, 2015. 3

[26] TorchVision maintainers and contributors. TorchVision: PyTorch's Computer Vision library, 2016. https://github.com/pytorch/vision, Commit:0ab7d05. 7

[27] Ran Meng, Zhengang Lv, Jianbing Yan, Gengshen Chen, Feng Zhao, Linglin Zeng, and Binyuan Xu. Development of spectral disease indices for southern corn rust detection and

severity classification. *Remote sensing*, 12(19):3233, 2020. 3

[28] Mengjun Miao and Shanqin Wang. Pa-colornet: progressive attention network based on rgb and hsv color spaces to improve the visual quality of underwater images. *Signal, Image and Video Processing*, 17(7):3405–3413, 2023. 3

[29] Trishen Munisami, Mahess Ramsurn, Somveer Kishnah, and Sameerchand Pudaruth. Plant leaf recognition using shape features and colour histogram with k-nearest neighbour classifiers. *Procedia Computer Science*, 58:740–747, 2015. 5, 6

[30] Albert Henry Munsell. *A basic treatise on the Color System of Albert H. Munsell: a Grammar of Color*. Van Nostrand Reinhold Company, 1969. 2

[31] Katherine M Murphy, Ella Ludwig, Jorge Gutierrez, and Malia A Gehan. Deep learning in image-based plant phenotyping. *Annual Review of Plant Biology*, 75, 2024. 1

[32] Ernest Mwebaze, Timnit Gebru, Andrea Frome, Solomon Nsumba, and Jeremy Tusubira. icassava 2019 fine-grained visual categorization challenge. *arXiv preprint arXiv:1908.02900*, 2019. 5, 6

[33] Shubham Nain, Neha Mittal, and Madasu Hanmandlu. Cnn-based plant disease recognition using colour space models. *International Journal of Image and Data Fusion*, 15(3):373–386, 2024. 3

[34] Nathalie Pettorelli. *The normalized difference vegetation index*. Oxford University Press, USA, 2013. 3

[35] Hafiz Tayyab Rauf, Basharat Ali Saleem, M Ikram Ullah Lali, Muhammad Attique Khan, Muhammad Sharif, and Syed Ahmad Chan Bukhari. A citrus fruits and leaves dataset for detection and classification of citrus diseases through machine learning. *Data in brief*, 26:104340, 2019. 5, 6

[36] Frank Rosenblatt. The perceptron: a probabilistic model for information storage and organization in the brain. *Psychological review*, 65(6):386, 1958. 2

[37] Arnout C Ruifrok, Dennis A Johnston, et al. Quantification of histochemical staining by color deconvolution. *Analytical and quantitative cytology and histology*, 23(4):291–299, 2001. 3, 6

[38] Neha Sharma and Mohamed Hefeeda. Hyperspectral reconstruction from rgb images for vein visualization. In *Proceedings of the 11th ACM multimedia systems conference*, pages 77–87, 2020. 1, 3

[39] Oskar Söderkvist. Computer vision classification of leaves from swedish trees, 2001. 5, 6

[40] JING WEI TAN and Siow-Wee Chang. D-Leaf. 2018. 5, 6

[41] William Wharton and Douglas Howorth. *Principles of television reception*. Pitman, 1967. 2

[42] Stephen Gang Wu, Forrest Sheng Bao, Eric You Xu, Yu-Xuan Wang, Yi-Fan Chang, and Qiao-Liang Xiang. A leaf recognition algorithm for plant classification using probabilistic neural network. In *2007 IEEE international symposium on signal processing and information technology*, pages 11–16. IEEE, 2007. 5, 6

[43] He Zhu, Xinyuan Wang, Tianxue Sun, Yanfei Yang, Chenhao He, Dong Chen, Li Zhang, Tao Ma, Yanxia Xing, Fuqiang Li, et al. Computer vision-based beef grading for marbling beef based on rgb-hsv color model and python software. In *2023 3rd International Signal Processing, Communications and Engineering Management Conference (ISPCEM)*, pages 609–614. IEEE, 2023. 3